

Mini Project: Bank Fraud Detection using Decision Tree + SMOTE (ninth week)

The goal of this mini project is to apply the decision tree algorithm to a real-world, significant problem in the financial industry: bank fraud detection.

Project Objective

You are required to build a machine learning model that classifies bank transactions into two categories:

0 → Normal Transaction

1 → Fraudulent Transaction

EDA and Visualization

- Load the Fraud Detection dataset (creditcard.csv)

Task EDA-1: Print dataset shape and first 5 rows

Task EDA-2: Check missing values

Task EDA-3: Plot class imbalance (Fraud vs Normal)

Task EDA-4: Explore the distribution of Amount.

Deliverable: Write 5–7 lines summarizing what. You discovered it from EDA.

Task EDA-5: Explain why accuracy is not reliable for this dataset. **Task EDA-5:** Split data (Stratified)

Task EDA-5: Handle rows with NaN values in the target column 'Class.'

Task 8: Baseline Decision Tree (without SMOTE)

Task 9: Apply SMOTE + Decision Tree

Write an explanation of why SMOTE is not a good practice with decision trees, and explain the results

Task 10: Draw Area Under Curve.

Task 11: Try to do hyperparameter tuning.